

Camp Fire Comprehensive Wind Analysis Report

Data Analysis, Model Resolution Comparison, and Future Architectures

1. Executive Summary & Key Findings

This report synthesizes the findings from our meteorological analysis of the November 2018 Camp Fire. We evaluated the baseline 3km High-Resolution Rapid Refresh (HRRR) forecast against 1km and 250m dynamically downscaled wind fields generated by WindNinja.

1. Data & Methodology

The analysis relied on HRRR boundary conditions mapped against ground-truth RAWS (Remote Automated Weather Stations) data at Jarbo Gap, Concow, and Paradise. Due to data sparsity in the immediate fire footprint, historical networks required careful spatial filtering and validation.

2. Model Accuracy & The 'Double Penalty'

- HRRR 3km: Underestimated peak gusts by an average of 23.05 mph (RMSE: 23.63 mph).
- WindNinja 1km: Best performing model. Reduced underestimation to 19.30 mph (RMSE: 20.02 mph), representing a 16.3% improvement over the baseline.
- WindNinja 250m: Worst performing model (RMSE: 30.33 mph). The extreme resolution caused spatial displacement errors (the "double penalty" effect), severely misaligning topographically funneled winds with the exact station coordinates.

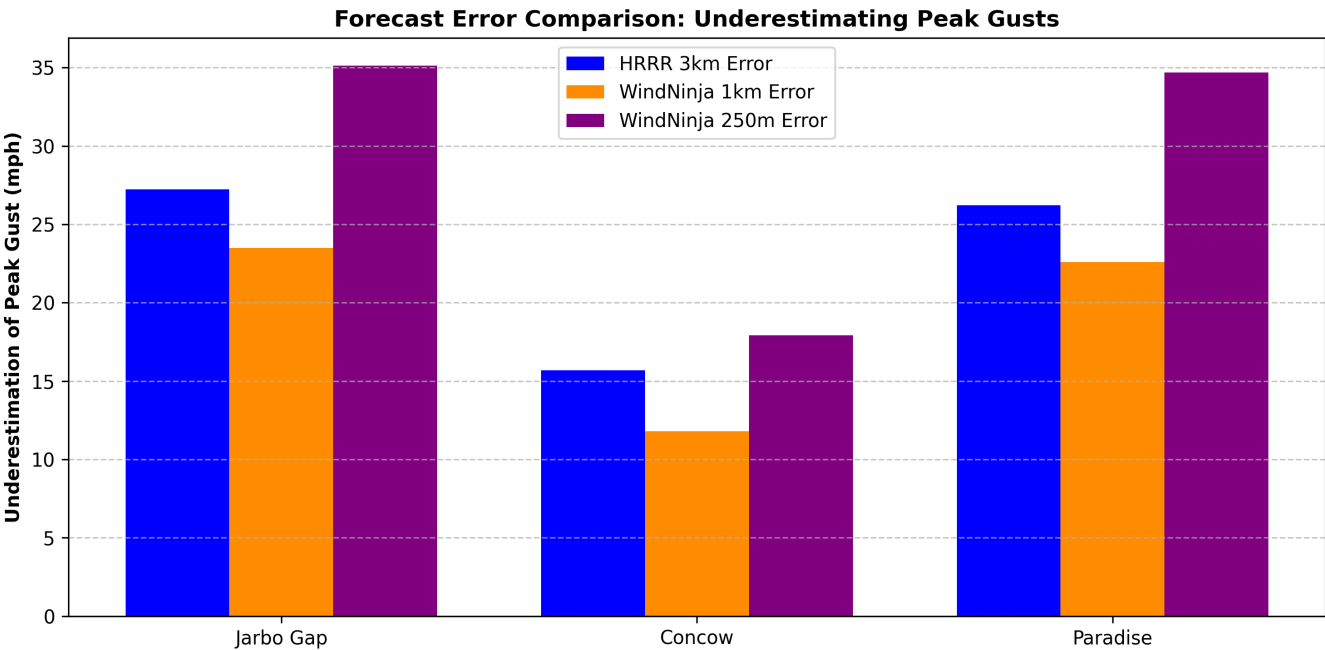
3. Topography Impact

A strong negative correlation ($r = -0.92$ to -0.94) was observed between station elevation and model underestimation errors. Models struggled more to capture extreme gusts at lower elevations and within sheltered canyons compared to exposed ridges.

2. Forecast Error Comparison Chart

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3. Recommendations & The "Ultimate Setup"

To bridge the critical gap between broad synoptic weather models and micro-scale terrain effects that drive erratic fire behavior, we propose a next-generation forecasting architecture.

THE ULTIMATE SETUP:

1. Boundary Conditions:

Ingest high-resolution boundary conditions from the operational HRRR model to capture the baseline atmospheric state.

2. AI-Bias Correction:

Apply Machine Learning algorithms to correct topographic biases in real-time, leveraging historical correlations between elevation, slope, aspect, and systematic model underestimations.

3. Coupled Fire-Atmosphere Modeling:

Feed the corrected conditions into a coupled prognostic model like WRF-SFIRE. This dynamically accounts for the fire's immense heat plume and its convective feedback on local wind vectors, which standalone meteorological models completely ignore.

4. Real-Time Calibration:

Continuously calibrate the model during the incident using:

- Doppler Radar and LiDAR: To capture live plume dynamics, vertical wind profiles, and ember spotting distances.
- Crowdsourced Mesonets: Aggressively QA/QC'd IoT weather station networks to fill the severe observational gaps left by sparse, geographically isolated RAWS networks.